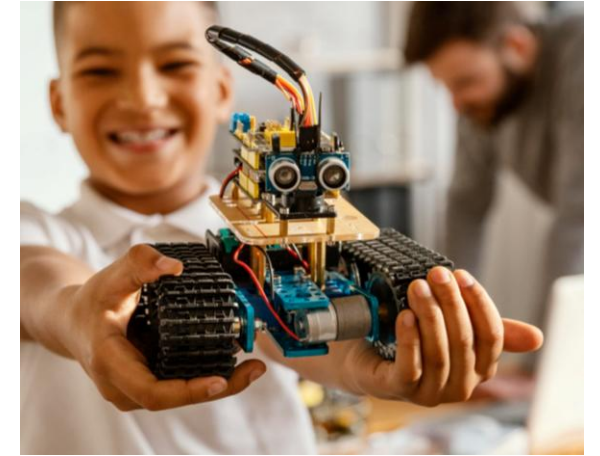
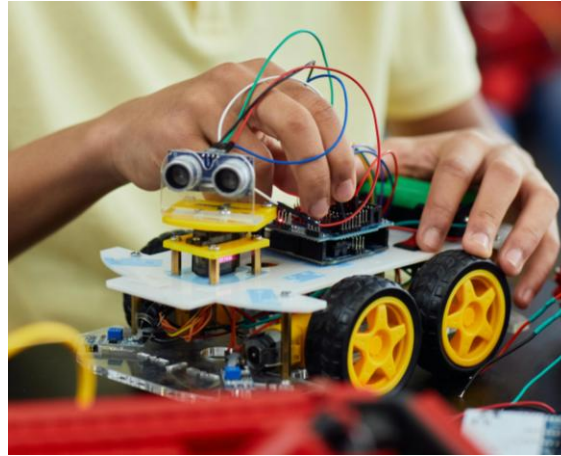
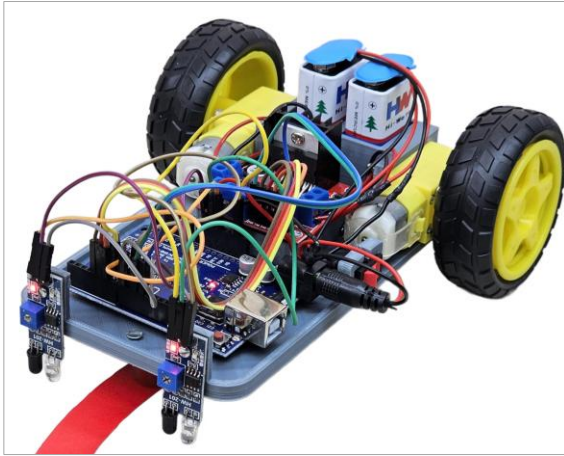


Arduino & Robotics

Smart Building Workshop



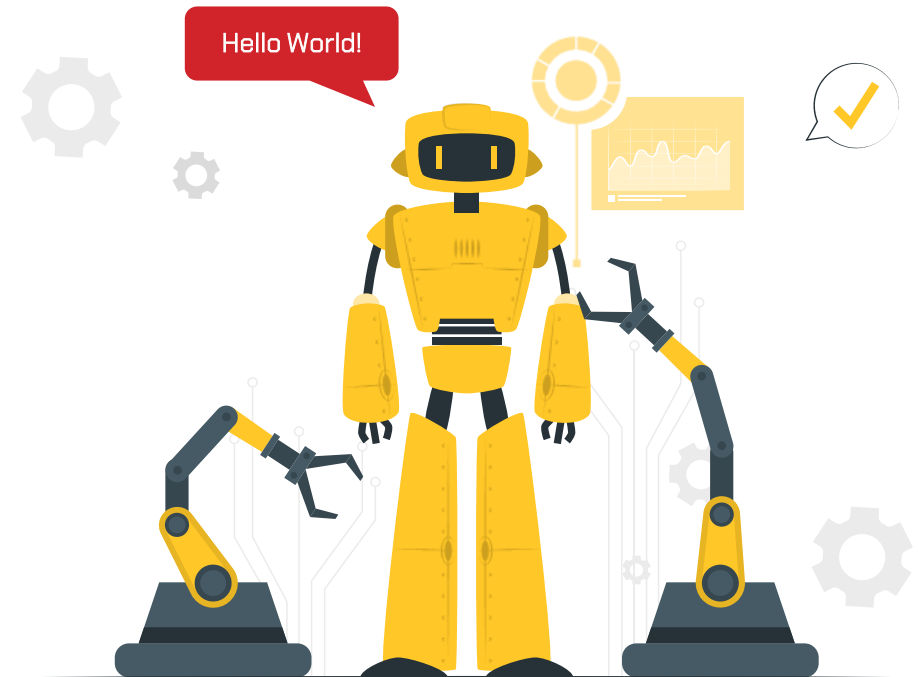
Hello World!

What is Robot?

A programmable machine that can complete a task

What about Robotics?

The field of study focused on developing robots and automation



Our Plan

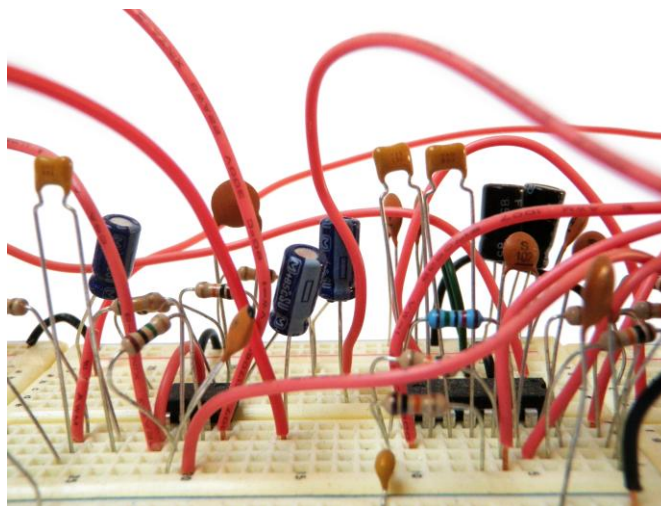
Day 1

- Ice-breaker & Introduction to Basic Electronics
- Arduino & Microcontroller Boards
- Computational Thinking
- Sensor Types Integration & Coding
- Setting up the bot – *Base Car*

Day 2

- Bot 1: Line Follower bot – *Beginner level*
- Bot 2: Obstacle Avoidance bot – *Intermediate level*
- Bot 3: Follow me bot – *Intermediate level*
- Project showcase





Introduction to **Basic Electronics**

What is ELECTRICITY?



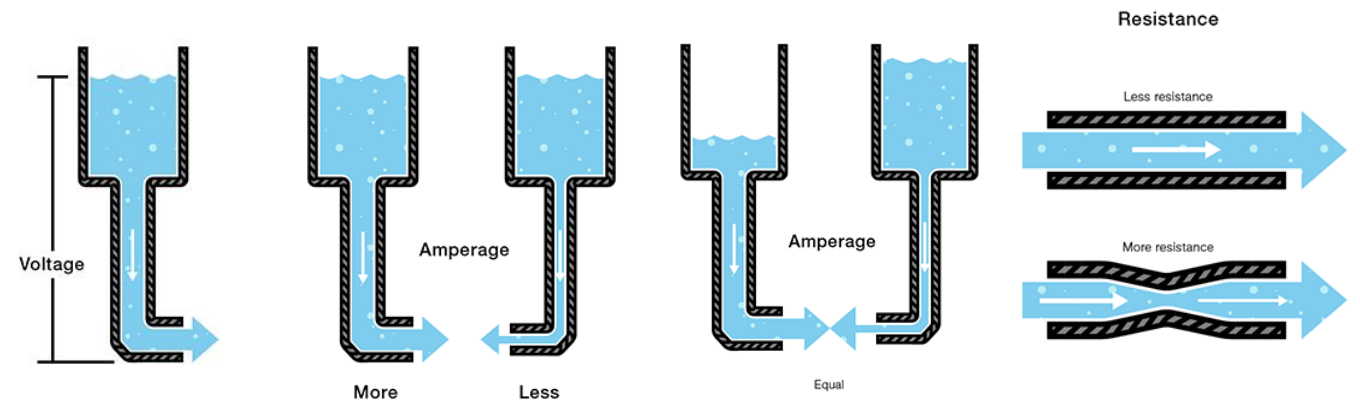
Electricity is the movement of electrons, creating charge, which we can harness to do work.

Electronics is the study of electrons, their behavior, flow, and control in different conditions, such as in gases, vacuums, or semiconducting materials

These are the three basic building blocks required to manipulate and utilize electricity

- **Voltage**
- **Current**
- **Resistance**

Imagine water flowing through pipes

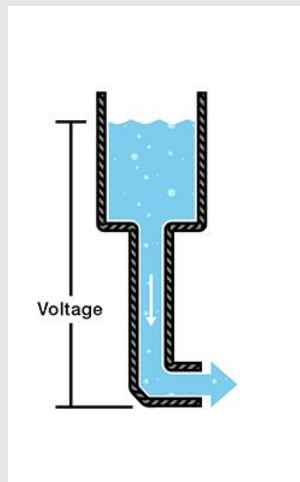
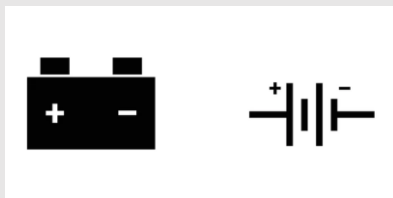


What is VOLTAGE?

The amount of electric potential between two points on a circuit.

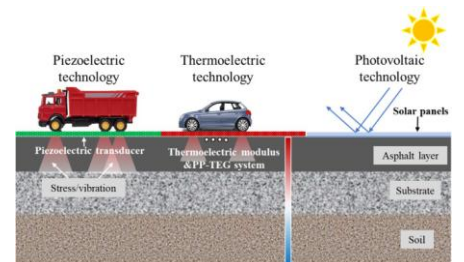
Derived unit for measuring voltage is the **volt** (V).

The conventional symbol for voltage is V



Voltage between points can be caused by the build-up of

- **Electric charge**
e.g. Capacitor
- **Electromotive force**
e.g. Generator, Turbines
- **Electrochemical processes**
e.g. Cells/Batteries
- Piezoelectric & Thermoelectric Effect

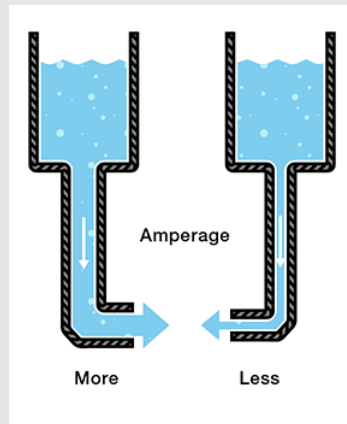
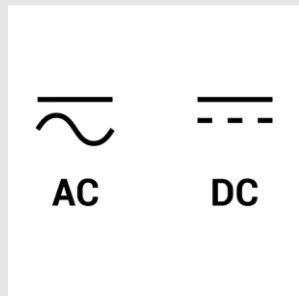


What is CURRENT?

Current is the rate at which electrons flow past a point in a complete/closed electrical circuit.

An **ampere** or *amp* (A), is the international unit used for measuring current.

The conventional symbol for current is I (for intensity)

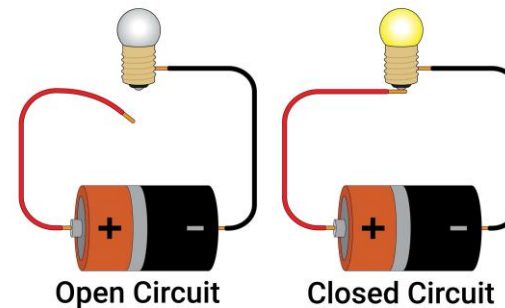


Electrons flow through a conductor when two pre-requisites of an electric circuit are met:

- The circuit includes an energy **source** (e.g., battery)



- The circuit forms a **closed**, conducting loop

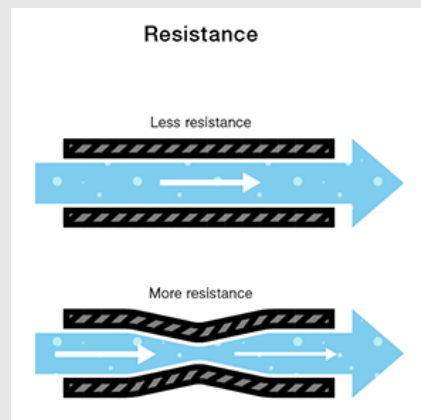
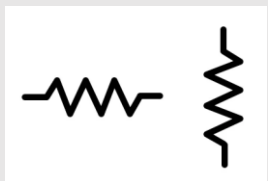


What is RESISTANCE?

The electrical **resistance** of an object is a measure of its opposition to the flow of electric current.

The SI unit of electrical resistance is the **ohm** (Ω).

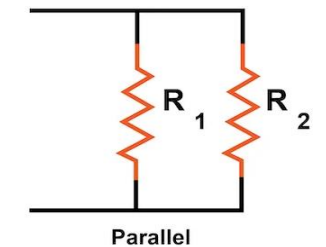
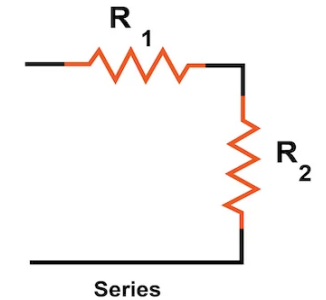
The conventional symbol for resistance is R



Types of Resistors



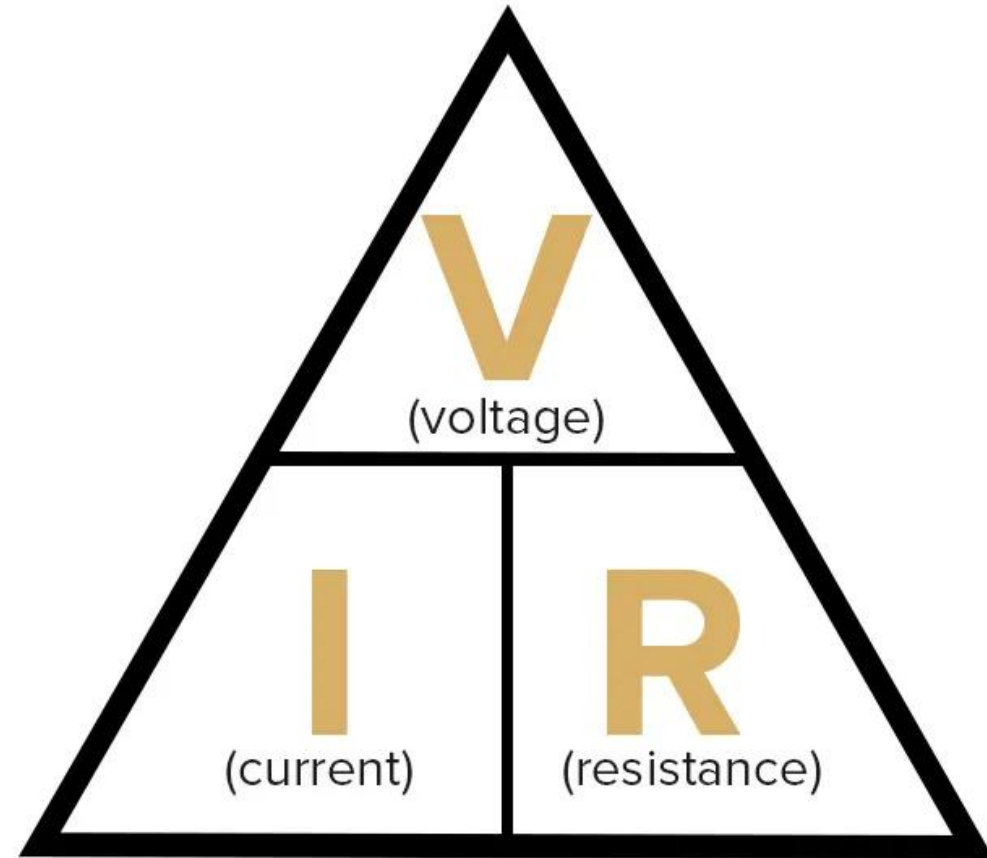
Connection types



OHM'S LAW & and Basic Circuit Principles

Ohm's law states that the electric **current** through a conductor is **directly** proportional to the **voltage** across it, and **inversely** proportional to the **resistance**.

$$V = IR \quad \text{or} \quad I = \frac{V}{R} \quad \text{or} \quad R = \frac{V}{I}$$



Identifying & Using Basic Electronic Components

Arduino Kit

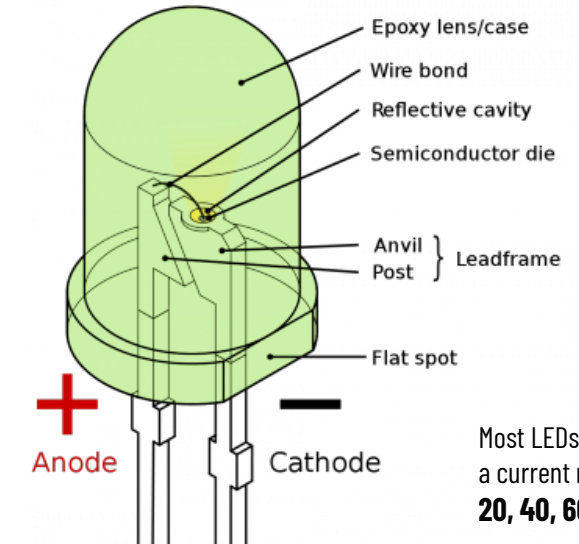
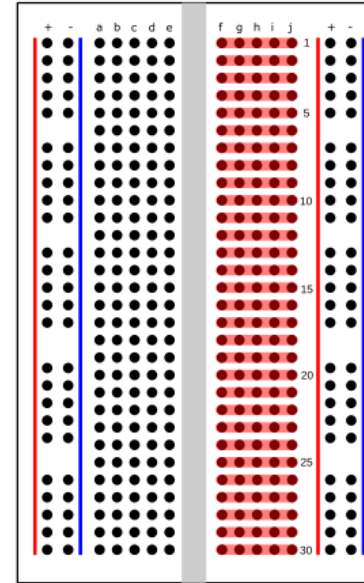


Hands-on Activity:

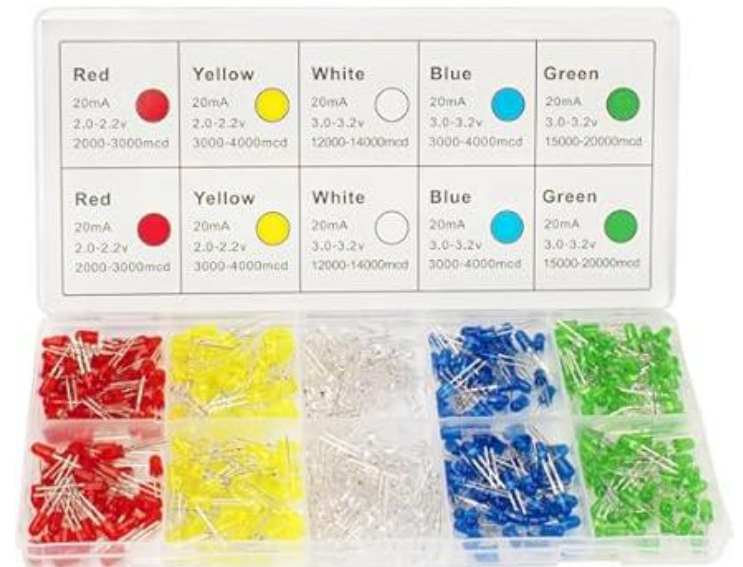
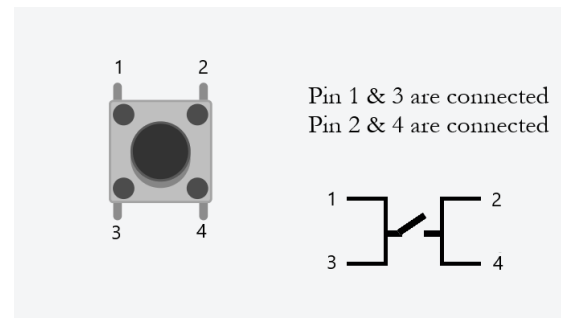
Constructing a basic LED circuit

Components Required:

- A breadboard
- An LED
- A resistor (? ohms)
- A button (optional)
- 9V battery and connector wires
- A multimeter

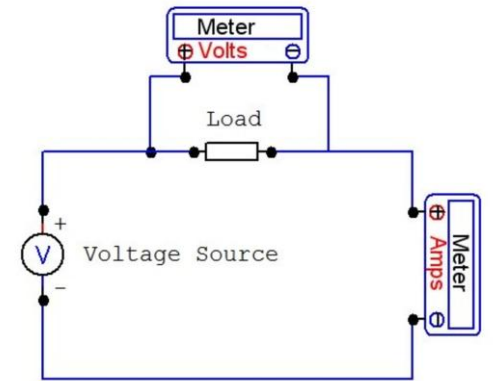


Most LEDs have a current rating of **20, 40, 60 or 80mA**



Hands-on Activity:

How multimeter works

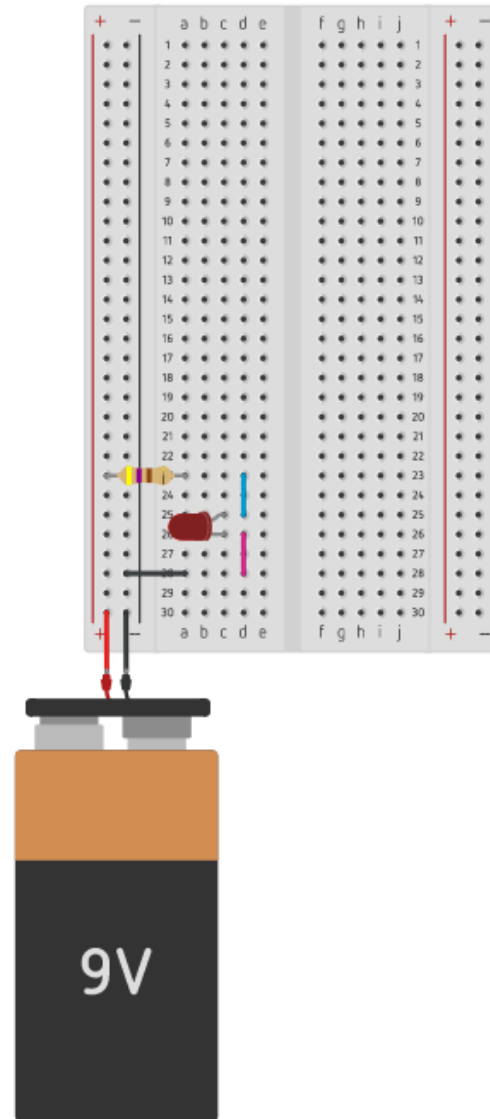


Hands-on Activity:

Constructing a basic LED circuit

Components Required:

- A breadboard
- An LED
- A resistor (? ohms)
- A button (optional)
- 9V battery and connector wires





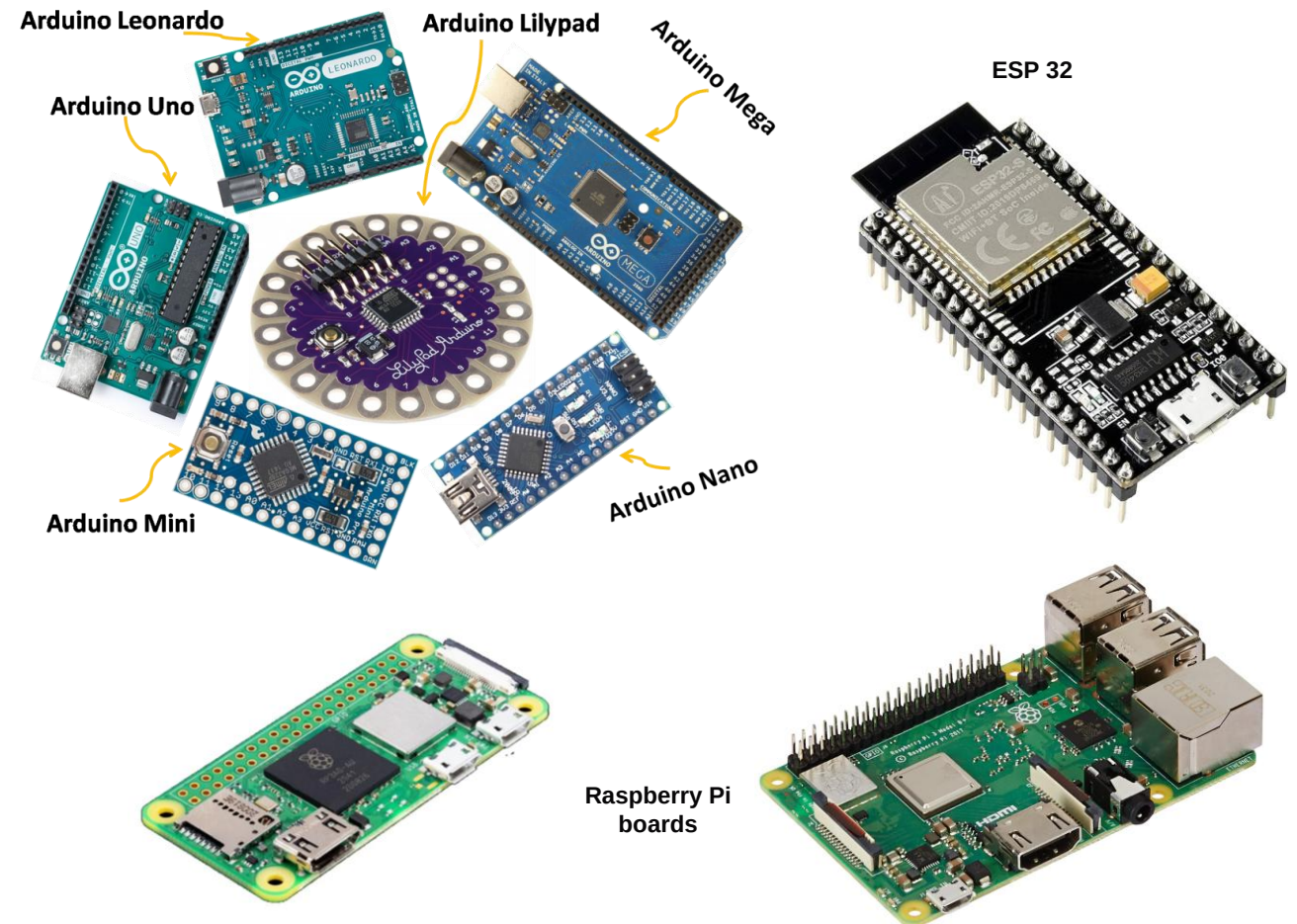
Introduction to **Arduino and Microcontroller Boards**

What is ARDUINO?

Arduino is an **open-source** electronics platform based on easy-to-use hardware and software. It's intended for anyone making interactive projects.

- They're user-friendly and perfect for beginners.
- Let's dive in!

Type of Arduino & micro-controller boards

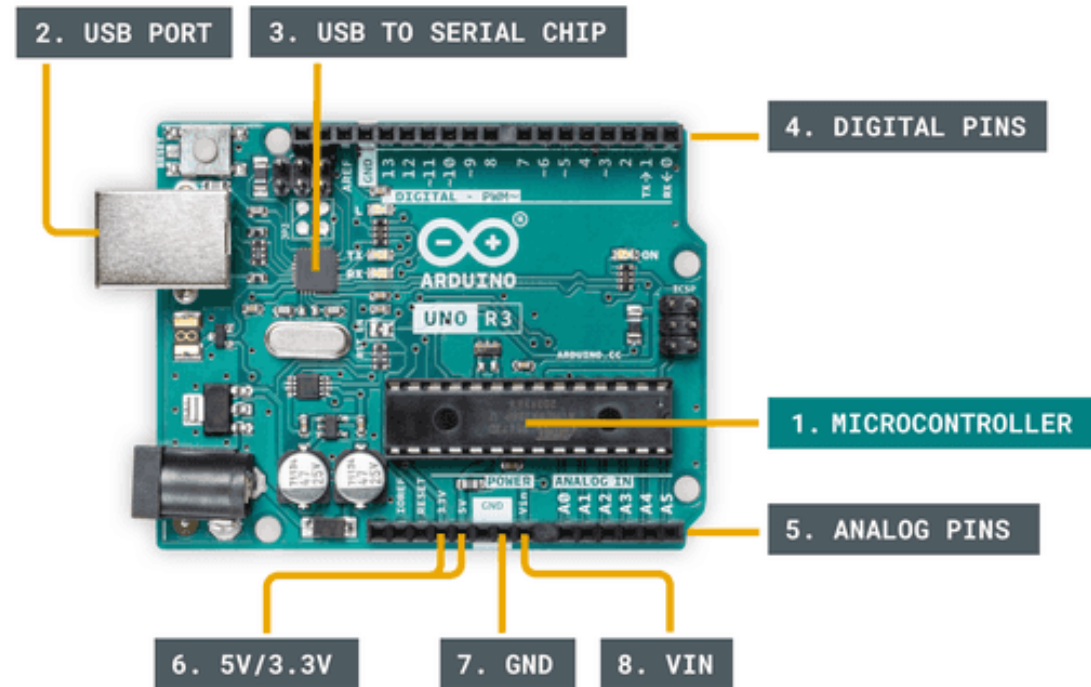


Arduino Uno

ANATOMY

Specifications

Microcontroller	ATmega328
Input Voltage Limit/Recommended	7-12V 6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
PWM Digital I/O Pins	6
Analog Input Pins	6
Flash Memory	32 KB (ATmega328P) of which 0.5 KB used by bootloader
EEPROM	1 KB (ATmega328)
SRAM	2 KB (ATmega328)
Clock Speed	16 MHz

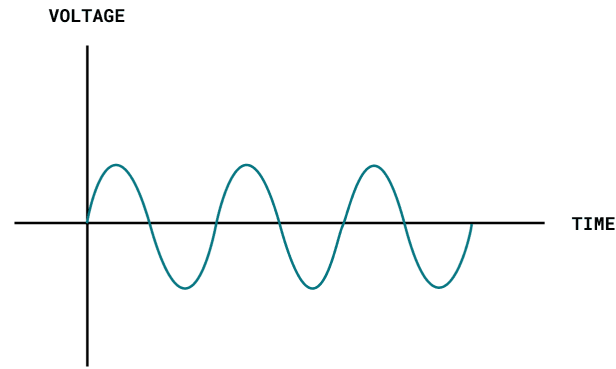


Electronic Signals

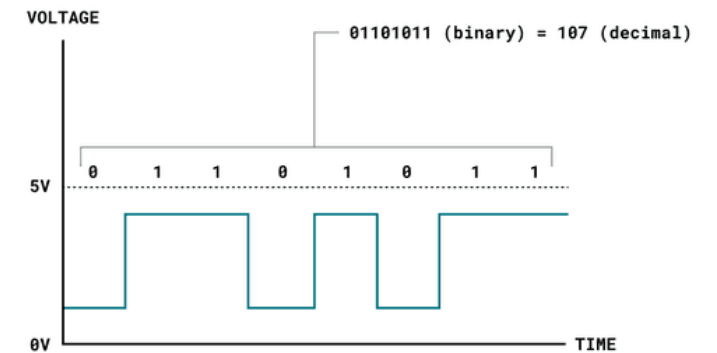
All communication between electronic components are facilitated by electronic signals. There are two main types of electronic signals: analog & digital.

Types:

- Analog Signals
- Digital Signals



An analog signal is generally bound to a **range**. In an Arduino, that range is typically 0 - 5v (or) 0 - 3.3v



A digital signal represents **only two binary states** (0 or 1) that are read as high or low states in the program.

Getting to know Arduino Software

Integrated Development Environment (Arduino IDE)

Things it can do:

- Manage boards, cores & libraries
- Write & upload programs
- Understanding Arduino code basic blocks



Skeleton of Arduino Program

How the program/sketch works

Things to note:

- Including Libraries
- Defining variables
- Setup & Loop function



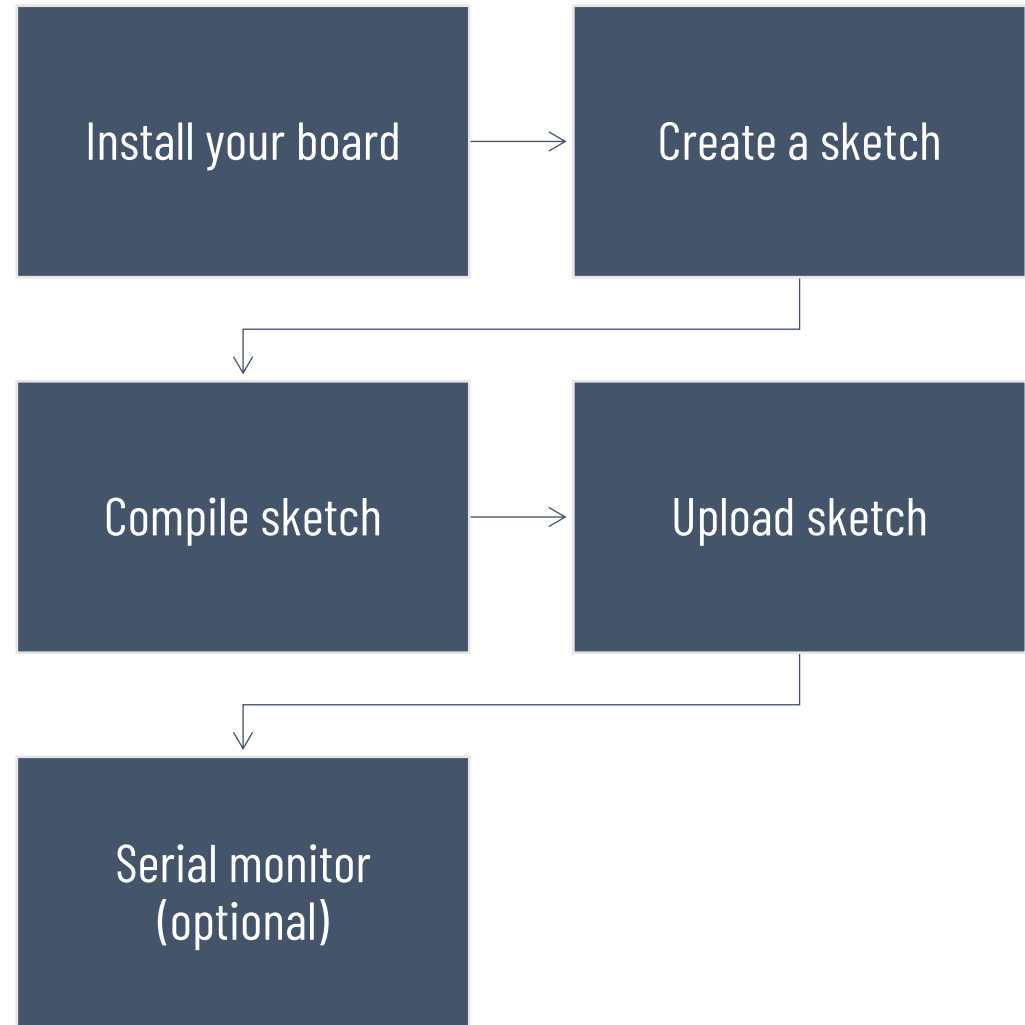
```
sketch_nov27a | Arduino IDE 2.3.3
File Edit Sketch Tools Help
Arduino Uno

sketch_nov27a.ino
1  /*
2  This is a comment at the top of a program,
3  it will not be recognized as code. Very good
4  to add an explanation of what your code does
5  here.
6
7  This sketch shows how to read a value from a
8  sensor connected to pin A1, print it out in
9  the Serial Monitor, and turn on an LED connected
10 to pin number 2 if a conditional is met.
11 */
12
13 int sensorPin = A1; //define pin A1 (analog pin)
14 int ledPin = 2; //define pin 2 (digital pin)
15 int sensorValue; //create variable for storing readings
16
17 //void setup is for configurations on start up
18 void setup() {
19   Serial.begin(9600); //initialize serial communication
20   pinMode(ledPin, OUTPUT); //define ledPin as an output
21 }
22
23 void loop() {
24   sensorValue = analogRead(sensorPin); // do a sensor reading
25
26   Serial.print("Sensor value is: "); //print a message to the serial monitor
27   Serial.println(sensorValue); //print the value to the serial monitor
28
29   //check if sensorValue is below 200
30   if(sensorValue < 200) {
31     digitalWrite(ledPin, HIGH); //if it is, turn on the LED on pin 2.
32   }
33   //if sensorValue is above 200, turn off the LED
34   else{
35     digitalWrite(ledPin, LOW);
36   }
37 }
```

Ln 37, Col 2 Arduino Uno on COM9 [not connected]

Typical Workflow of a Program

To upload code to an Arduino board using the IDE, one typically does the following

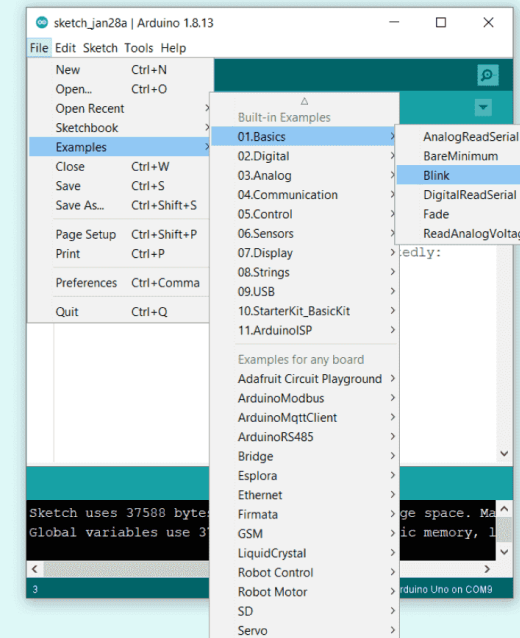
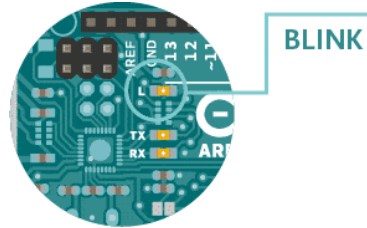


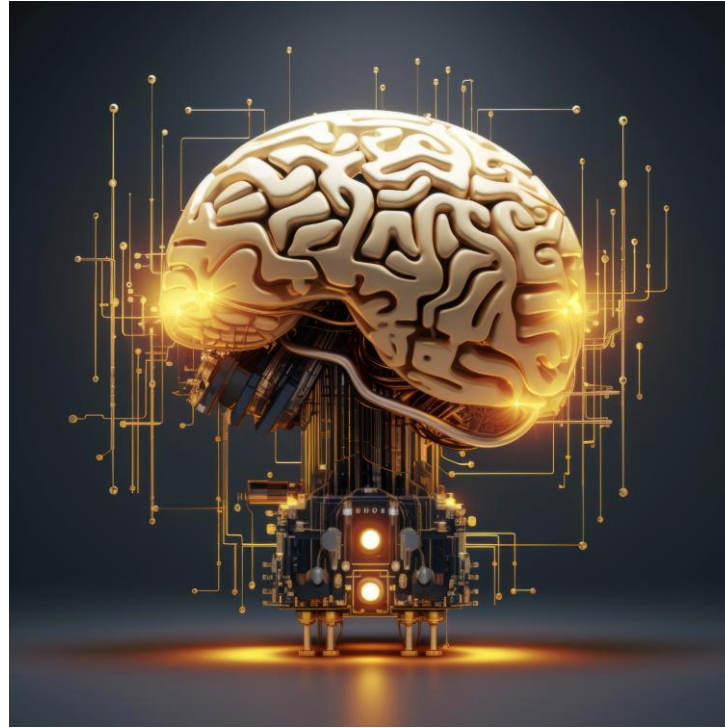
Your first Arduino sketch

Writing and Uploading a BLINK

Things to do:

- Setup IDE for **Arduino Uno** board
- Open Blink example from menu
- Preview code
- Upload & Test





Computational Thinking

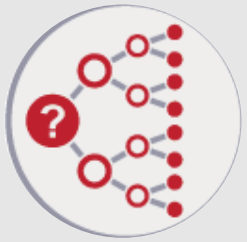
What is Computational Thinking?

Computational Thinking is a way of **solving problems, designing systems**, and understanding **human behavior** by drawing on the concepts fundamental to computer science.

- Enhances problem-solving skills
- Encourages logical thinking
- Prepares future careers in technology
- Promotes creativity and innovation



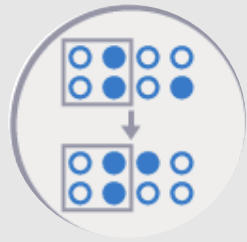
How it works?



Decomposition:

Breaking down a big problem into smaller, manageable parts

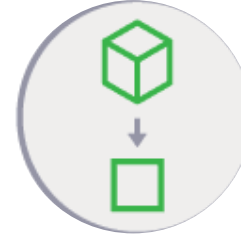
e.g., planning a birthday party by breaking it down into tasks like invitations, decorations, cake, & games



Pattern Recognition:

Finding similarities or patterns

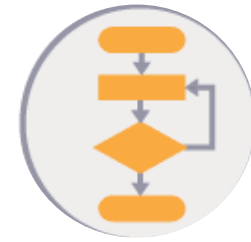
e.g., recognizing the pattern in a series of numbers or identifying common themes in stories



Abstraction:

Focusing on important information and ignoring irrelevant details

e.g., summarizing a story or simplifying a complex drawing



Algorithm Design:

Creating a step-by-step solution

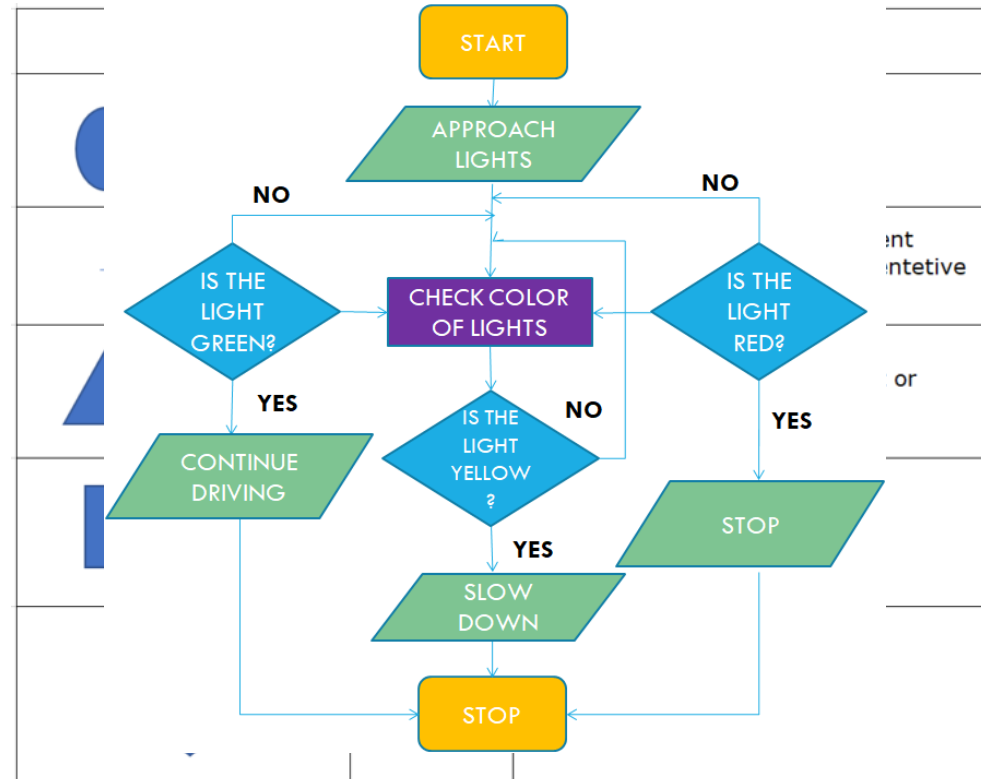
e.g., writing a simple recipe or instructions for a game

Flowchart

Represents a workflow or process.

A flowchart can also be defined as a **graphical or symbolic representation** of an algorithm.

Each step in the process is represented by a different symbol and contains a short description of the process step.





Sensor & Actuators Integration & Coding

What are Sensors & Actuators?

They are essential components in Arduino projects, bringing them to life by enabling interaction with the environment.

Sensors, is used to **sense** its environment, meaning it records a physical parameter.

Actuator, is used to **activate** or change a physical state.

Typically referred to as **inputs** and **outputs**.



Examples

Modules	Output / Input
A light (e.g., LED)	OUTPUT
Temperature Module	INPUT
A motor	OUTPUT
A switch	OUTPUT
Infrared Module	INPUT
Buzzer module	OUTPUT

How to interact?

With sensors and actuators

Things to note:

- Reading sensor values
- Controlling Actuators

Sensor readings

```
int sensorValue = analogRead(A0);  
int sensorValue = digitalRead(2);
```

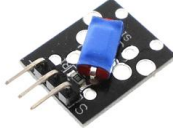
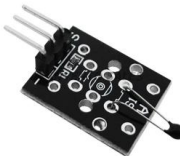
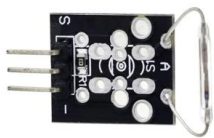
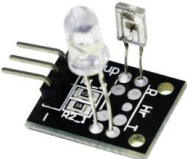
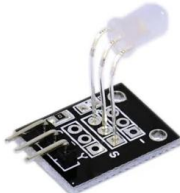
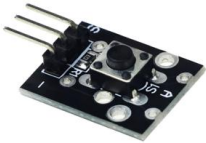
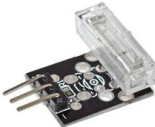
- First line reads the value from a sensor connected to **analog** pin A0 and stores it in the variable *sensorValue*
- Second line reads from **digital** pin A0

Actuator control

```
digitalWrite(13, HIGH);  
analogWrite(9, 255);  
servo.write(90);
```

- Turns on an LED connected to digital pin 13
- Sets the speed of a motor connected to PWM pin 9 to its maximum value (255)
- Setting servo motor to a 90-degree position

Sensor & Actuators: **Integration & Coding**

								
1 x PS2 Game JoyStick Module	1 x 1 Channel Relay 5V	1 x Sound Sensor Module	1 x Small Microphone Sound Sensor Module	2 x Magic Cup Light Module	1 x Mercury Medallion Module	1 x Tilt Switch Module	1 x Photosensitive Resistor Sensor Module	1 x Digital Temperature Module
								
1 x TCRT5000 Tracking Module	1 x IR Obstacle Avoidance Sensor Module	1 x The Flame Sensor Module	1 x Linear Magnetic Hall sensor	1 x DHT11 Temperature and Humidity Sensor Module	1 x Hall Magnetic Sensor Module	1 x DS18B20 Temperature Sensor Module	1 x Analog Temperature Module	1 x Active Buzzer Module
								
1 x Dry Reed Pipe Module	1 x Mini Dry Reed Pipe Module	1 x Finger Detecting Heartbeat Module	1 x 7 Color Flashing LED Module	1 x DIP 3 Color LED Module	1 x SMD 3 Color LED Module	1 x 5mm Two-Color LED Module	1 x 3mm Two-Color LED Module	1 x Passive Buzzer Module
								
1 x Laser Module	1 x Tactile Switch Module	1 x Vibration Switch Module	1 x Blue Rotary Encoder Module for Arduino with Demo Code	1 x Infrared Transmit Sensor Module	1 x Infrared Receiver Module for Arduino	1 x The Knock Sensor Module	1 x Broken Light Module	

DHT11

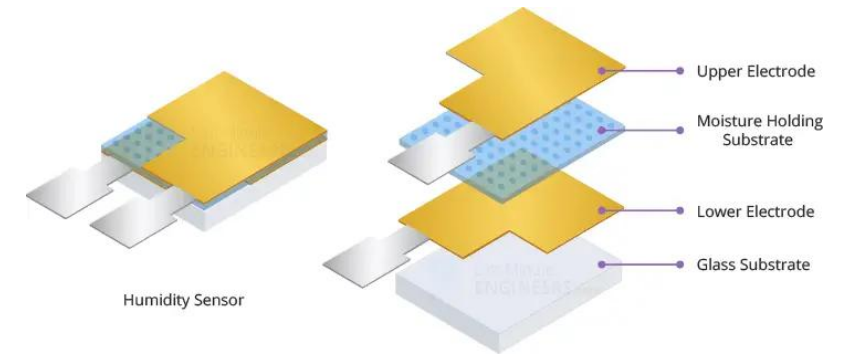
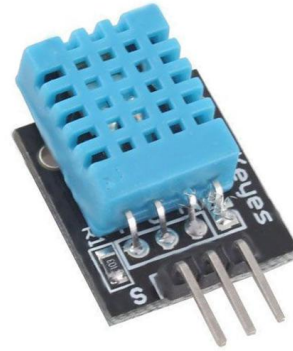
Digital Humidity & Temperature Sensor

Time to sense & act

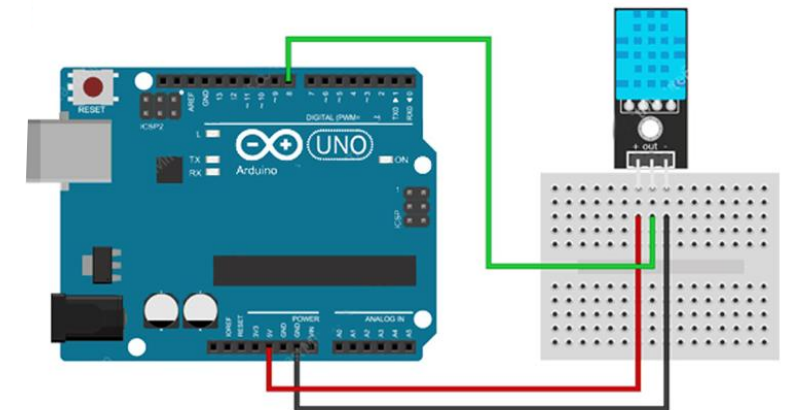
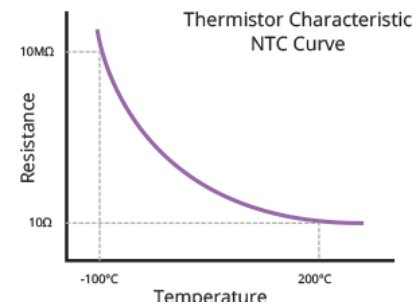
- Humidity measurement range: 20–95%
- Humidity measurement accuracy: $\pm 5.0\%RH$
- Temperature measurement range: $0\text{--}50^{\circ}\text{C}$
- Temperature measurement accuracy: $\pm 2.0^{\circ}\text{C}$
- Working voltage: 3.3–5V

External library

dhtlib



NTC Thermistor

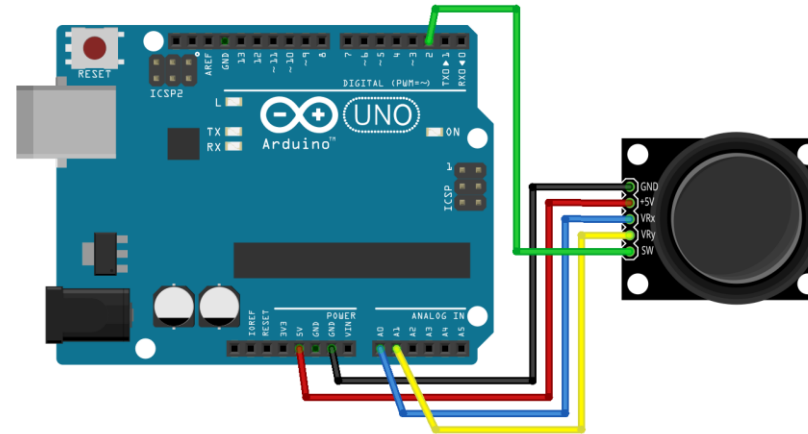
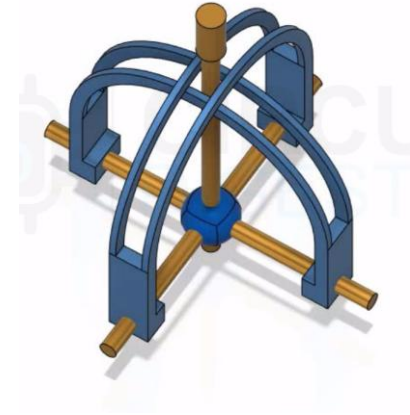


Joystick Module

PS2 (PlayStation 2) controllers

Measure position coordinates on the X and Y axis

- Operating Voltage: 5V
- A switch that is press-able by pushing the "hat" down

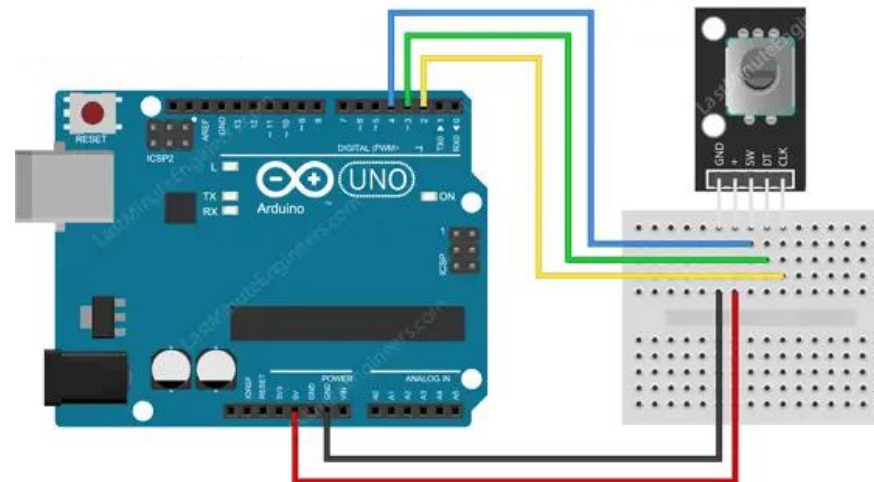
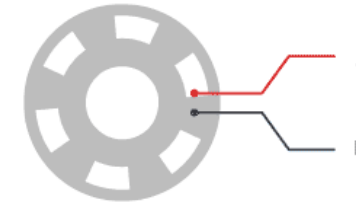
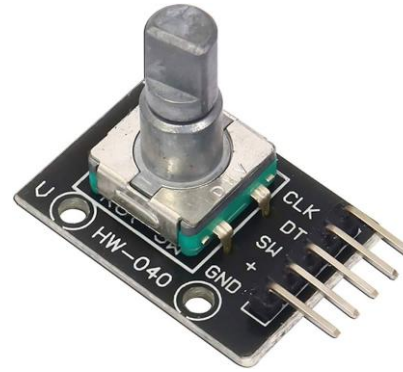


Rotary Encoder

A rotary input device (as in knob)

Indication of how much the knob has been rotated AND what direction it is rotating

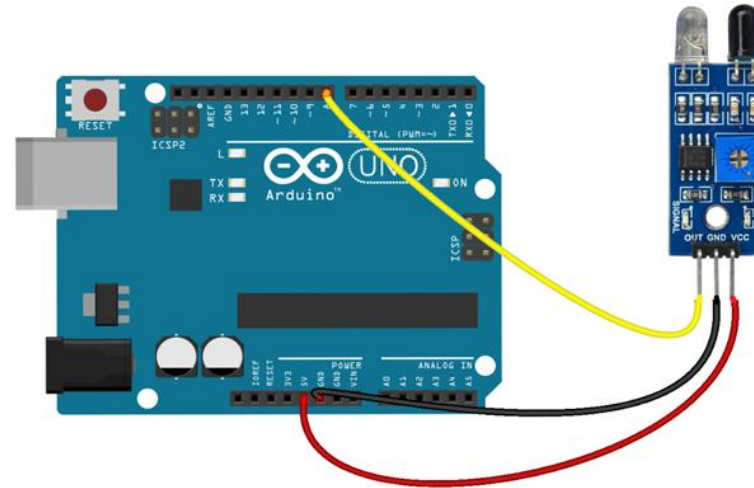
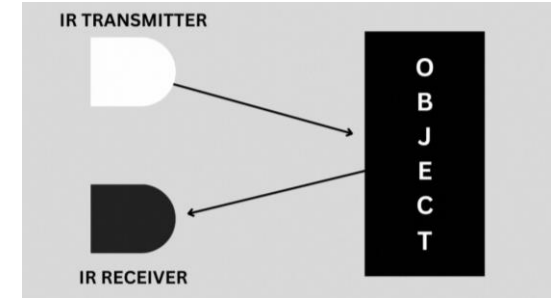
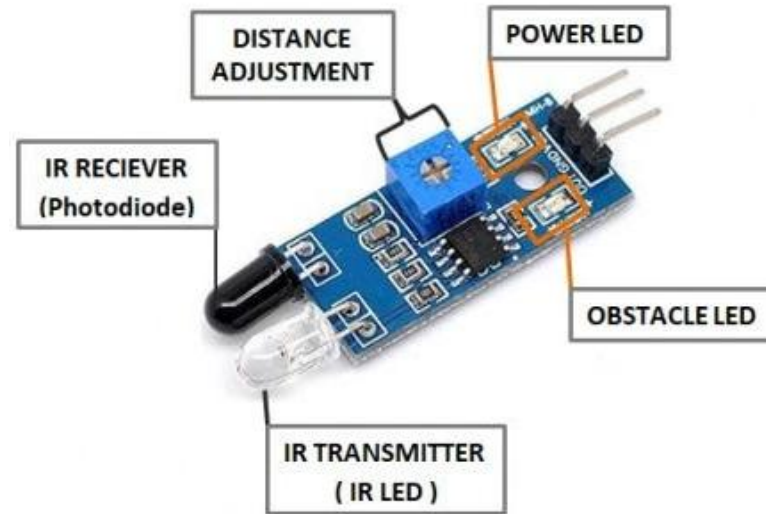
- Type : Incremental Encoder.
- Cycles per revolution (CPR) : 20.
- Working voltage : 3.3 – 5V



IR Detection Module

Time to sense & act

- Operating Voltage: 3.6 to 5 V
- Detection distance: 2 ~ 30cm
- Detection angle: 35°



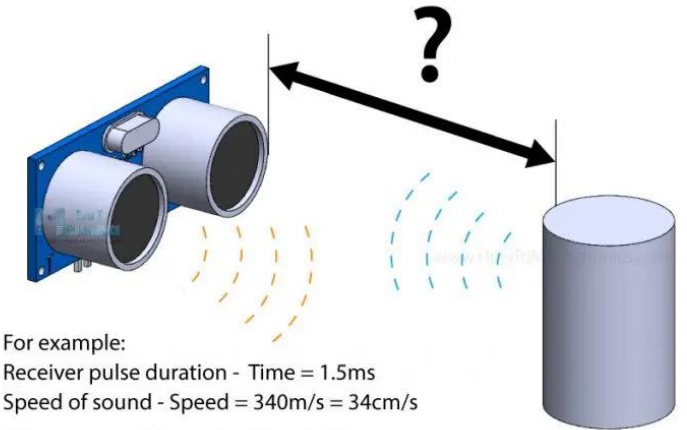
Ultrasonic Sensor

Time to sense & act

- Operating Voltage: 5 V
- Sonar Sensing Range: 2-400 cm
- Max. Sensing Range: 450 cm
- Measuring angle: 15 °
- Frequency: 40 kHz

External library

Newping (optional)



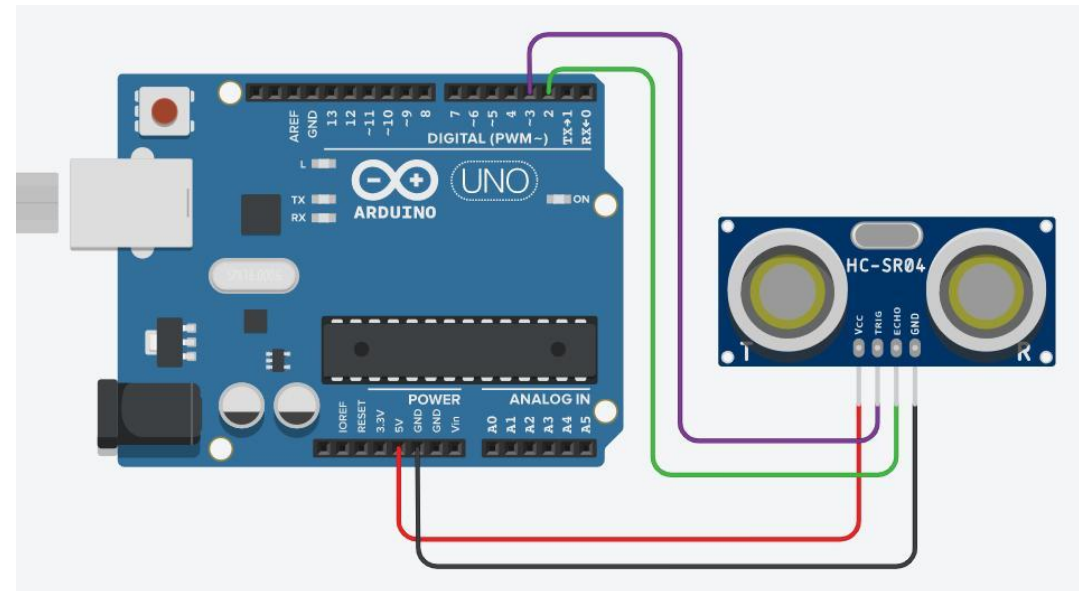
For example:

Receiver pulse duration - Time = 1.5ms

Speed of sound - Speed = 340m/s = 34cm/s

$$\text{Distance} = (\text{Speed} \times \text{Time}) / 2$$

$$\text{Distance} = (34\text{cm/ms} \times 1.5\text{ms}) / 2 = 25.5\text{cm}$$



Motor Driver

Specifications

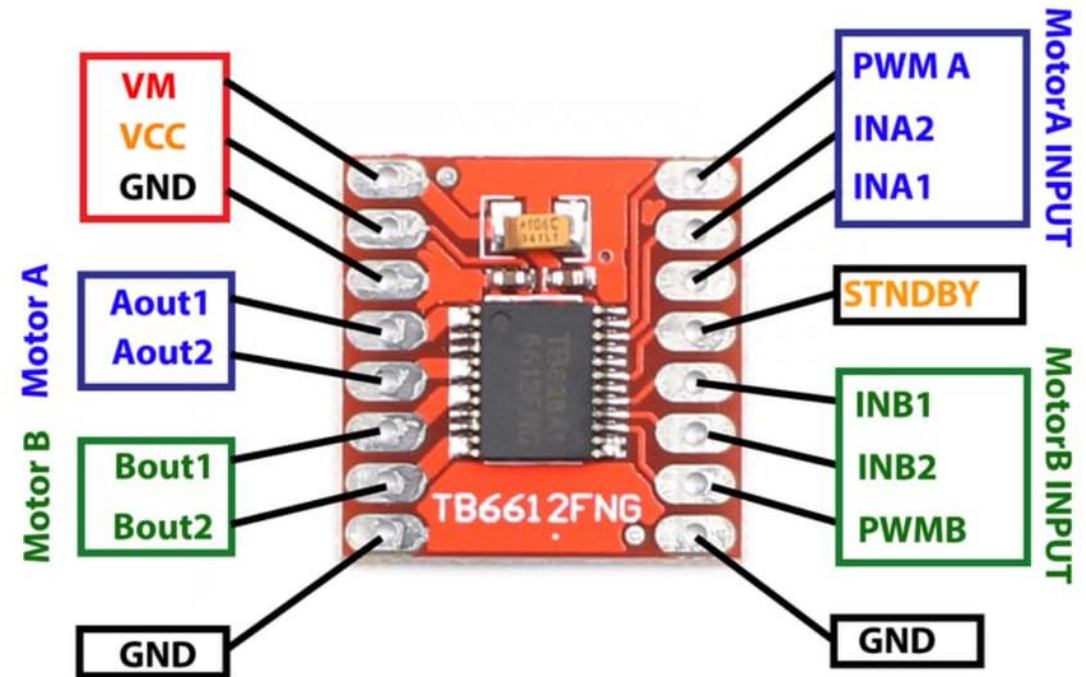
- **Control modes:** CW/CCW/short-break/stop-motor
- Built-in thermal shutdown circuit and a low voltage detecting circuit
- Standby control to save power

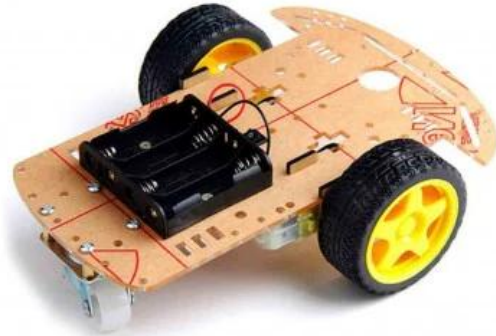
External library

```
#include <SparkFun_TB6612.h>
```

https://github.com/sparkfun/SparkFun_TB6612FNG_Arduino_Library

To find a library for your product, search for the product name or a keyword such as '**k type**' or '**digitizer**', and the library you want should show up.





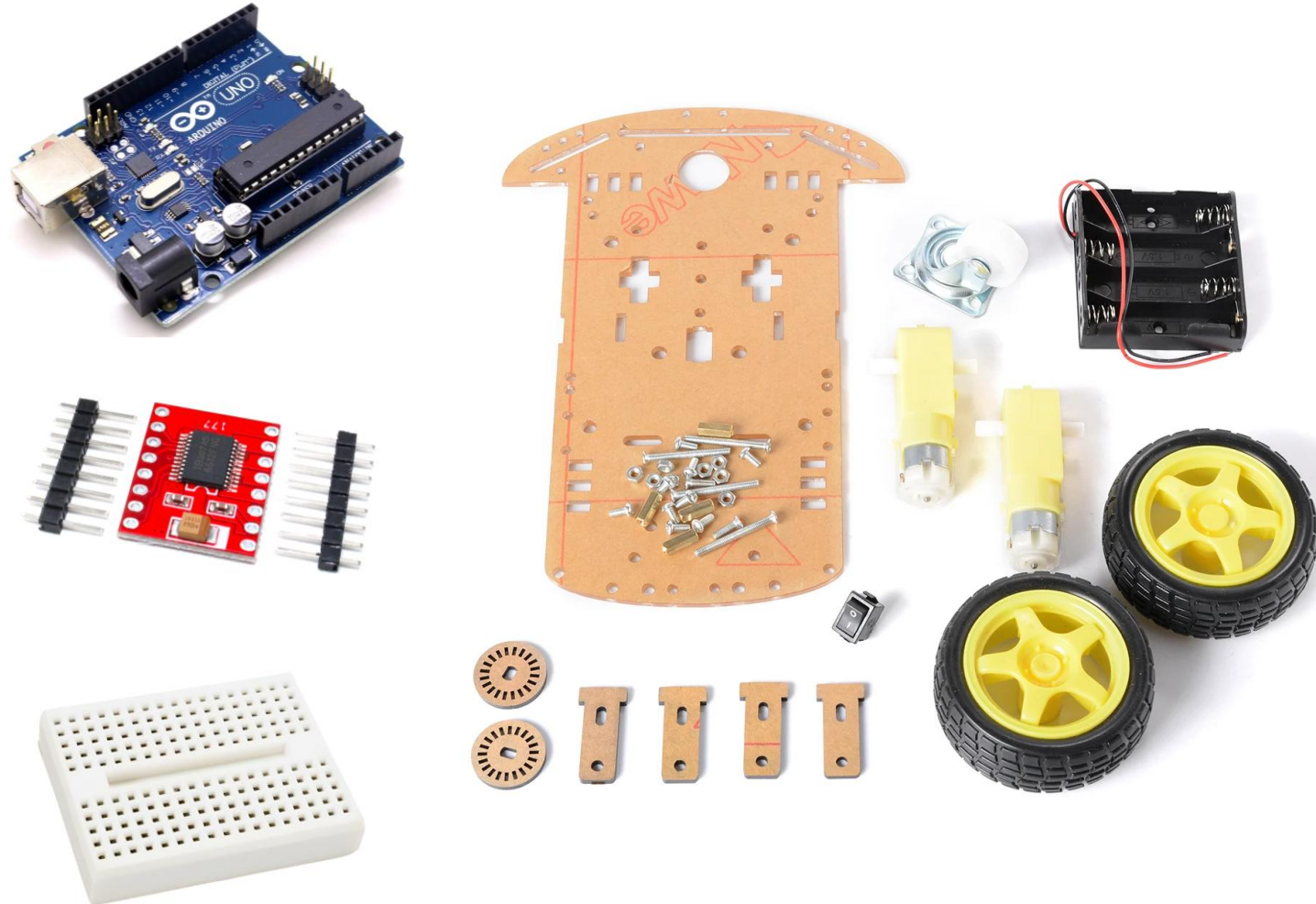
Setting up the Bot **Base Car**

Hands-on Activity: Step 1

Assembling the robot chassis

Components Required:

- Chassis Kit
- Motors & Wheels
- Arduino Uno
- Motor Driver, Breadboard
- Switch, Batteries & Holder

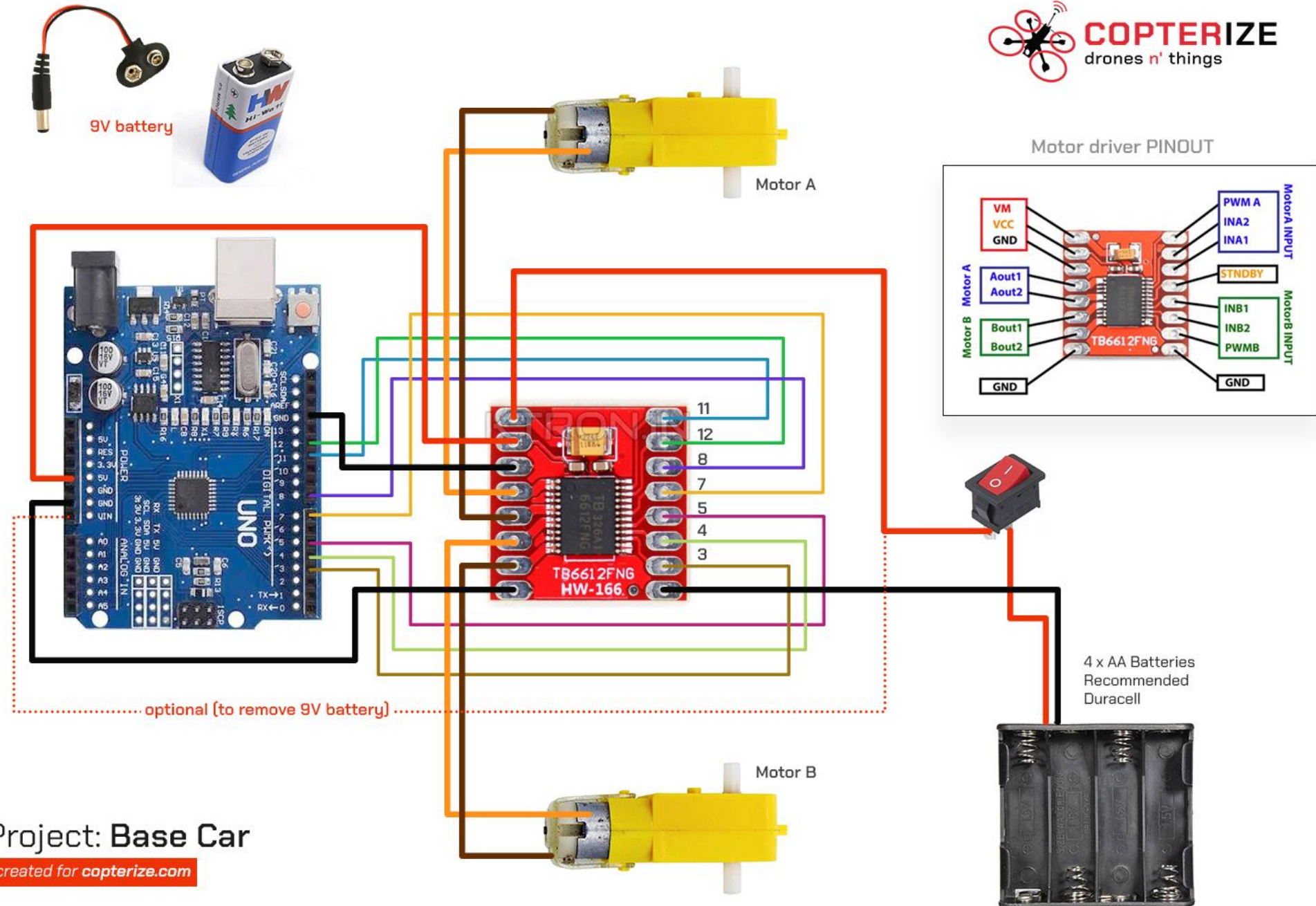


Hands-on Activity: Step 2

Circuit Diagram & Logic

Things to do:

- Solder motors
- Motor driver module connections
- Connecting to Arduino

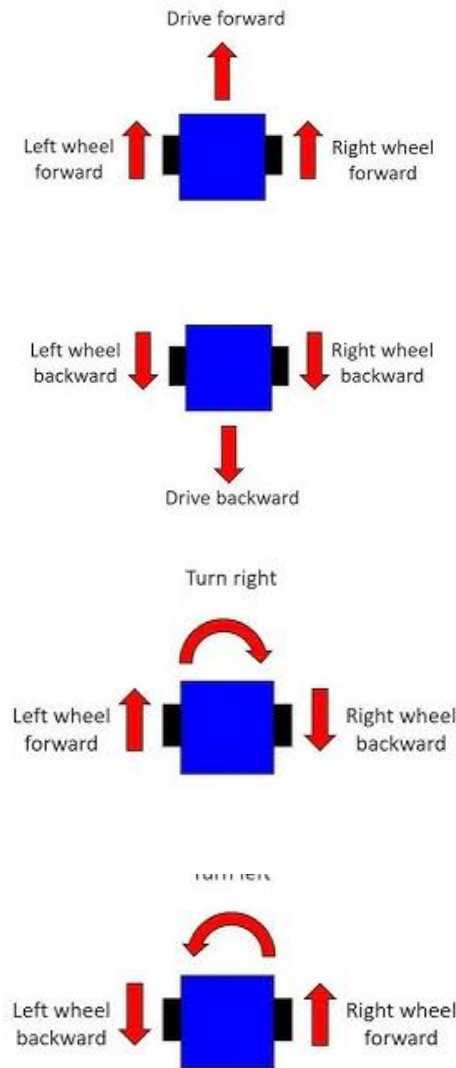


Hands-on Activity: Step 3

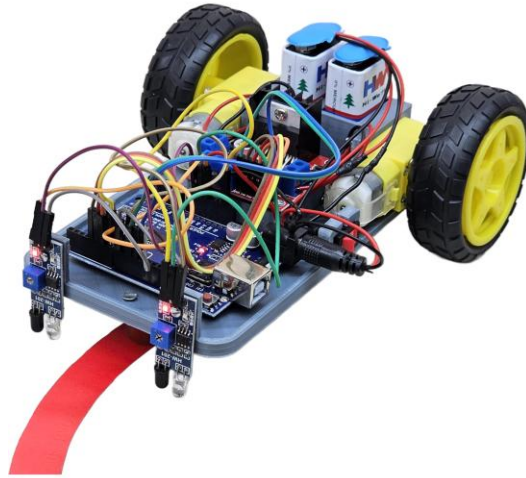
Programming & Testing

Things to do:

- Code & Test control motors
- Upload program
- Testing motor movement (Left, Right, Forward & Backward)



```
void fwd() {  
    forward(motor1, motor2, mvtSpeed);  
    Serial.print("FWD - ");  
}  
void bwd() {  
    back(motor1, motor2, -mvtSpeed);  
    Serial.print("BWD - ");  
}  
void stp() {  
    brake(motor1, motor2);  
    Serial.print("STOP - ");  
}  
void lft() {  
    left(motor1, motor2, turnSpeed);  
    Serial.print("LFT");  
}  
void rgt() {  
    right(motor1, motor2, turnSpeed);  
    Serial.print("RGT");  
}
```



Building Bot 1

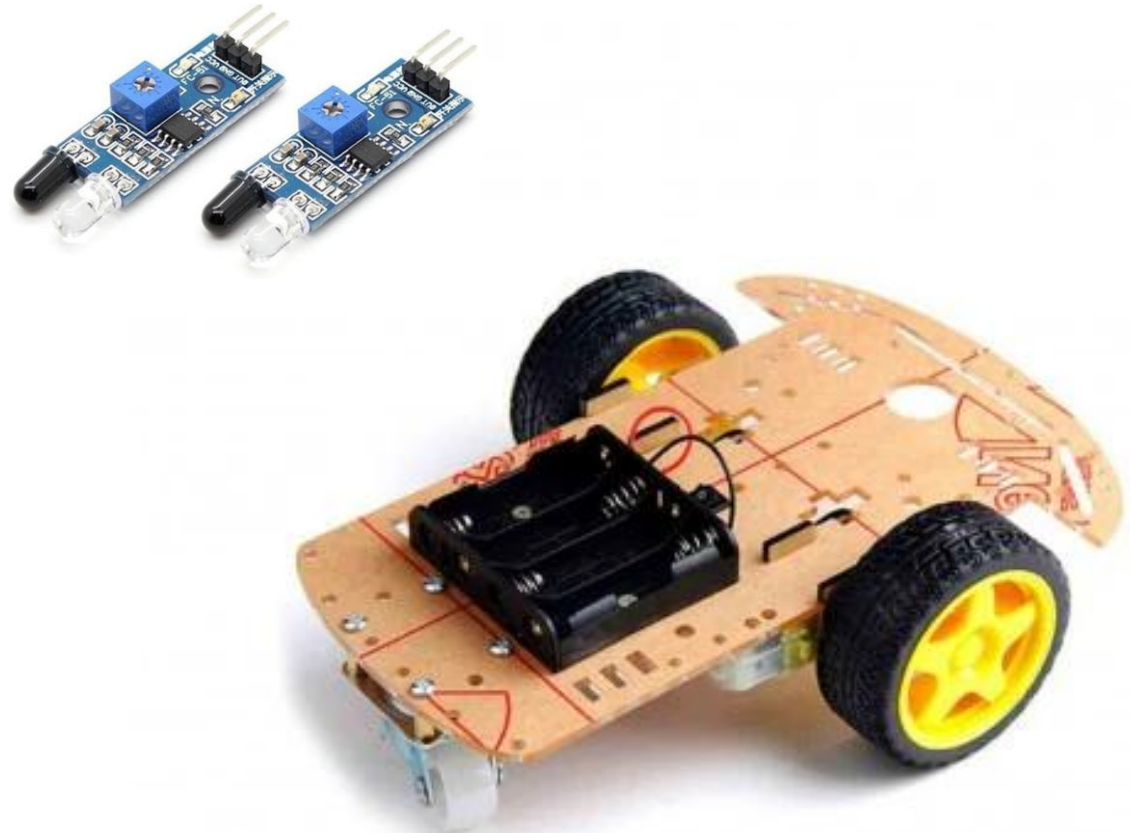
Line Follower

Hands-on Activity: Step 1

Assembling the robot

Components Required:

- Base car
- IR sensors
- & few wires

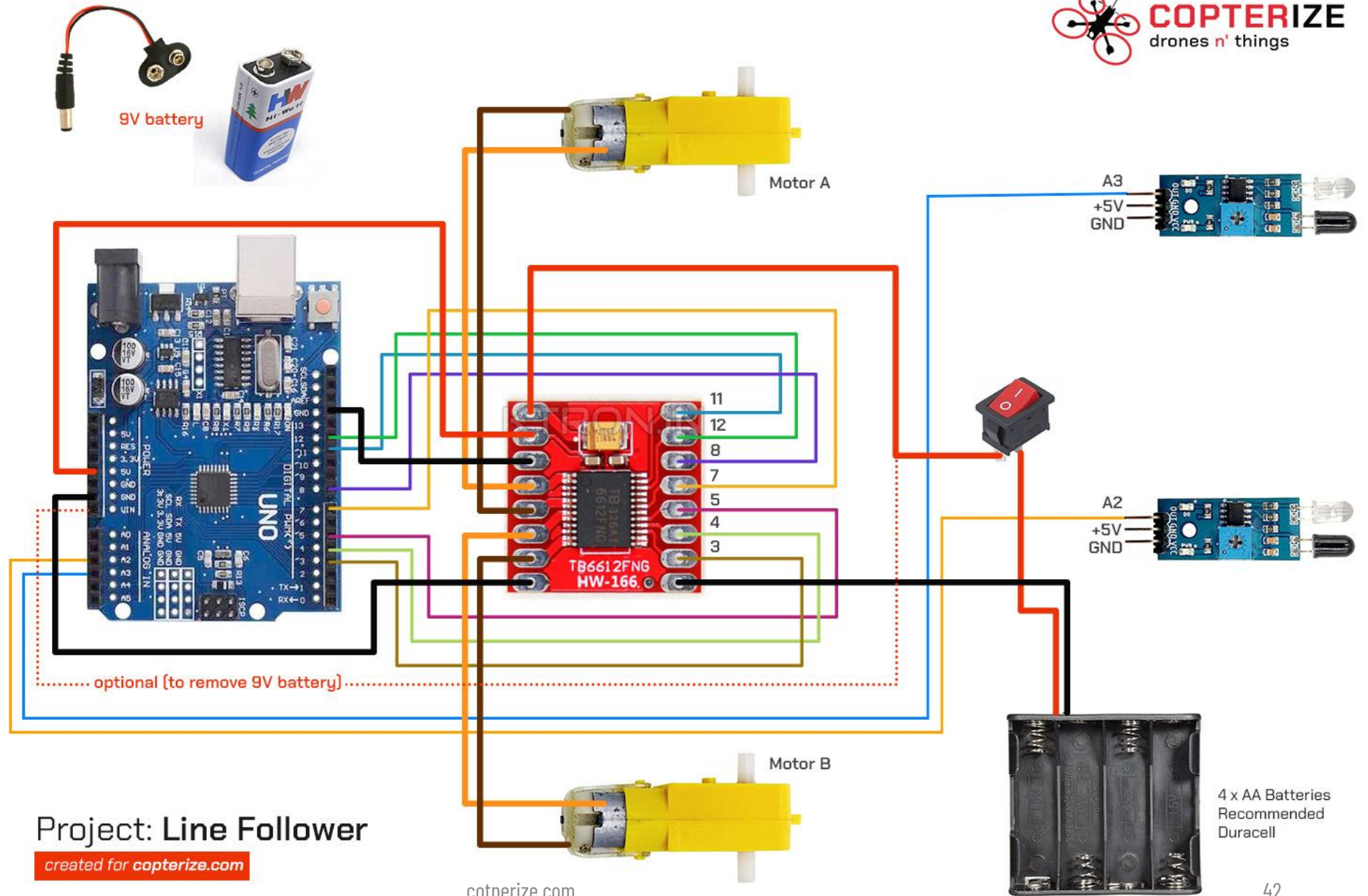


Hands-on Activity: Step 2

Programming Motor control

Things to do:

- Connect IR Sensor
- Code & test control motors
- Upload program
- Testing motor movement (Left, Right, Forward & Backward)





Building Bot 2

Sense & Avoid

Hands-on Activity: Step 1

Assembling the robot chassis

Components Required:

- Robot smart car
- Ultrasonic sensors
- Servo motor
- Sensor mounting bracket

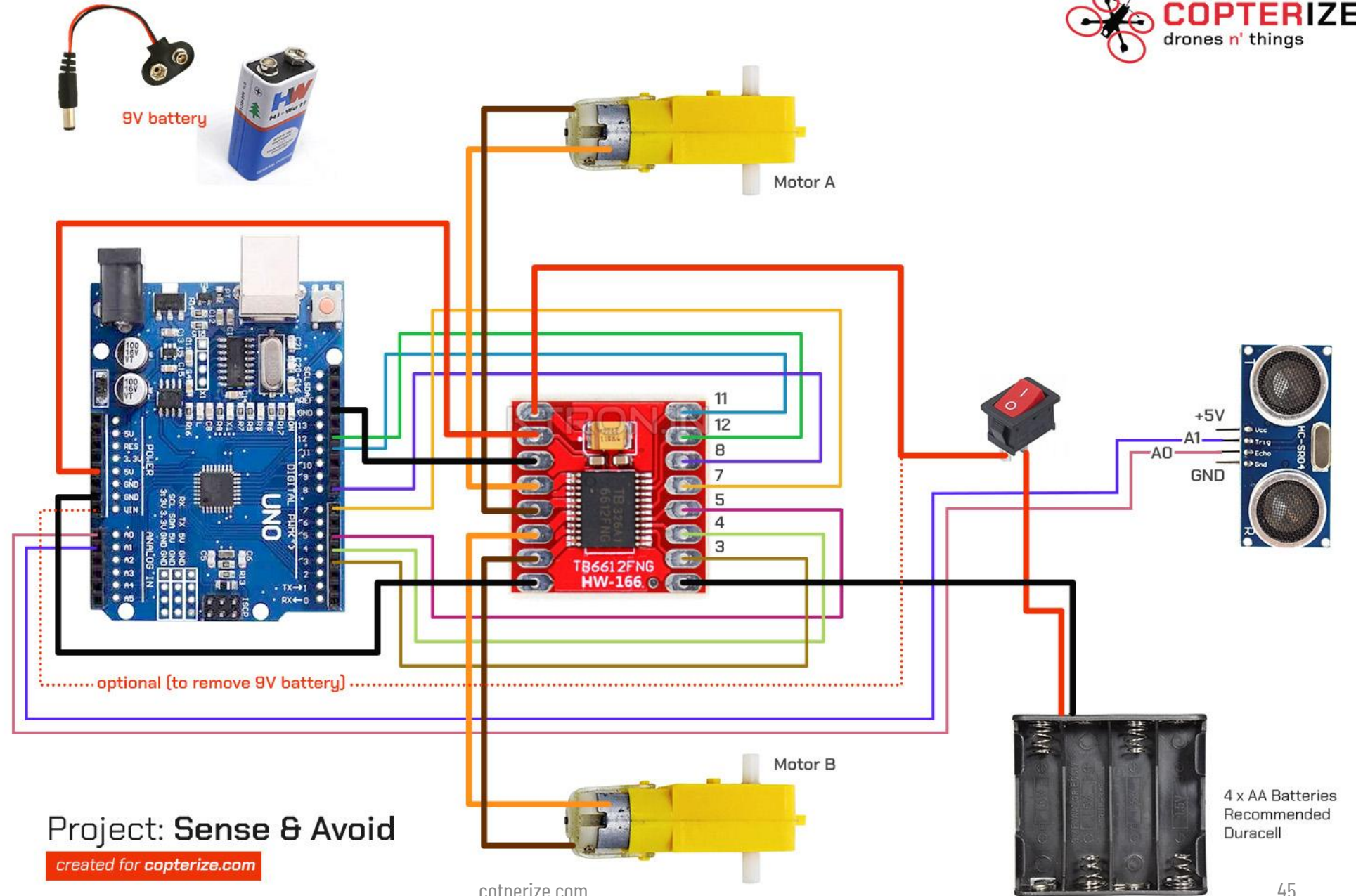


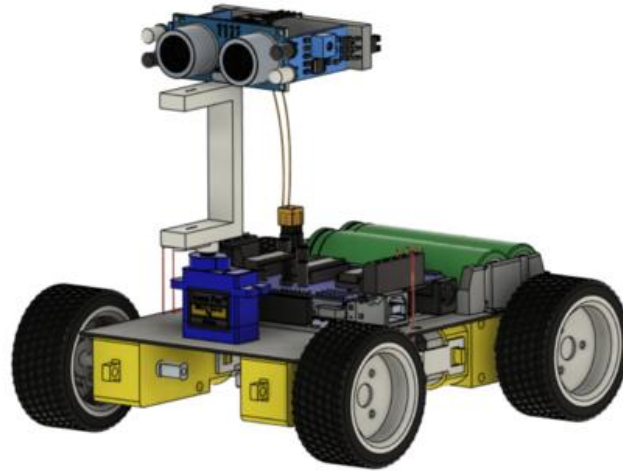
Hands-on Activity: Step 2

Programming Motor control

Things to do:

- Solder motors & Battery connections
- Code & test control motors
- Upload program
- Testing motor movement (Left, Right, Forward & Backward)





Building Bot 3

Follow me!

Hands-on Activity: Step 1

Assembling the robot chassis

Components Required:

- Robot smart car
- IR sensors
- Ultrasonic sensors
- Servo motor
- Sensor mounting bracket

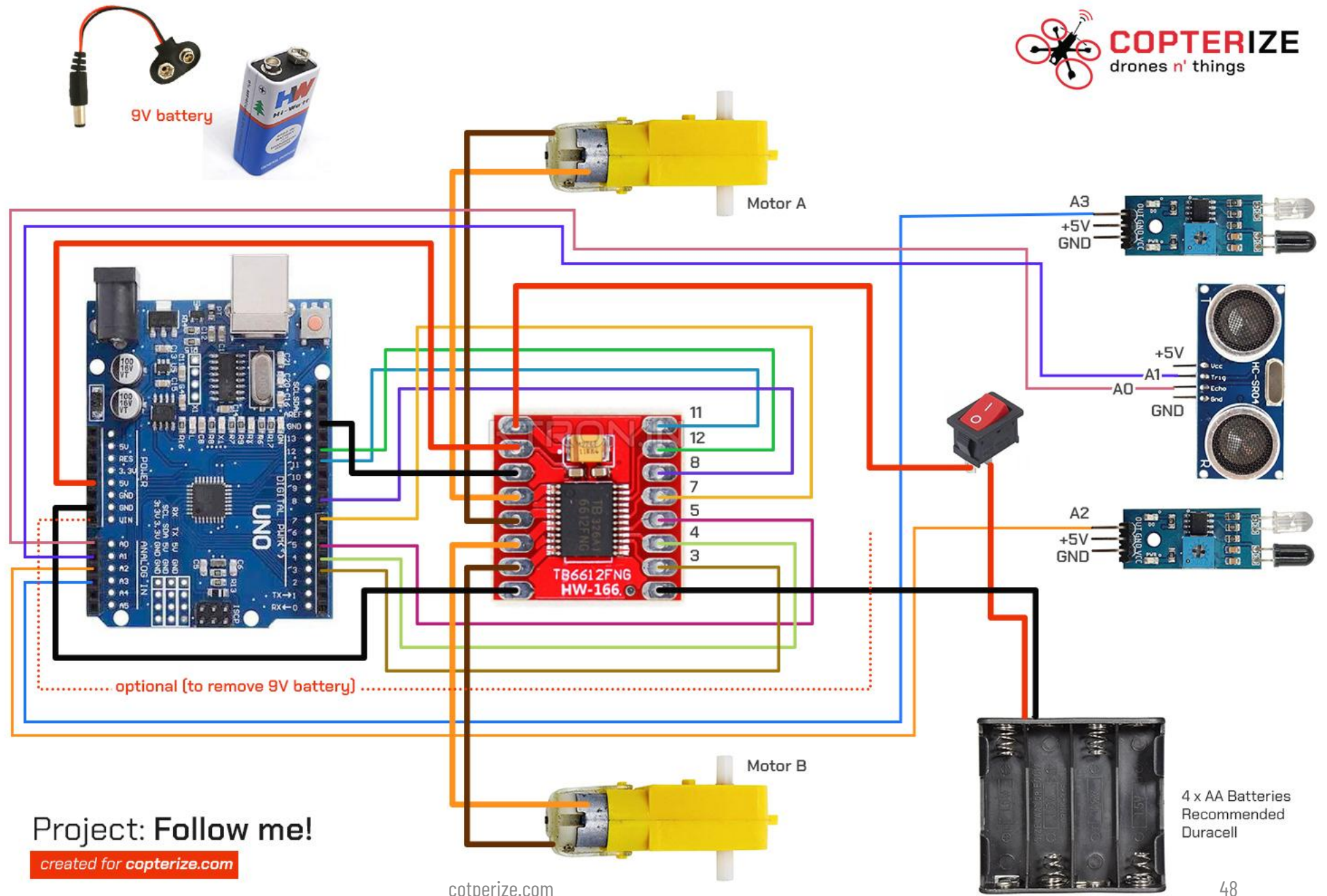


Hands-on Activity: Step 2

Programming Motor control

Things to do:

- Solder motors & Battery connections
- Code & test control motors
- Upload program
- Testing motor movement (Left, Right, Forward & Backward)



Final Projects & Showcase

Let's Innovate!

